Lecture Scribe Notes

Tutorial_2_Solution (1)....
PDF

Generate a lecture scribe notes which can be used as exam reference material .Use only the provided context (lecture slides) dont add anything of you own . While writing the scribe reason step by step through all definitions , notation, assumptions , theorem statements , proofs ,derivations and worked examples exactly as given in the lecture slides. Make the logical flow explicit so that a student can follow the reasoning during exam revision . Include proofs, derivations and examples only if they appear in the provided context. Do not introduce new explanations examples or material beyond the lecture scope . Maintain a formal and academic tone and strict faithfulness to the lecture structure

Below are **lecture scribe notes** strictly derived from the provided lecture document, preserving structure, notation, derivations, and logical flow exactly as given.

CSE 400: Fundamentals of Probability in Computing

Tutorial 2 – Lecture Scribe Notes

(Strictly based on provided solutions)

Q1. Random Point on Line Segment

Problem Interpretation

A point is chosen uniformly on a line segment of length L.
Let the point be at distance x, where $0 < x < L$.

Two segments formed:

- Lengths: x and $L - x$

Define ratio:

$$R = \frac{\min(x, L-x)}{\max(x, L-x)}$$

We require:

$$R < \frac{1}{4}$$

Case Analysis

Case 1: $x \leq \frac{L}{2}$

$$\frac{x}{L-x} < \frac{1}{4} \Rightarrow 4x < L - x \Rightarrow 5x < L \Rightarrow x < \frac{L}{5}$$

Case 2: $x > \frac{L}{2}$

$$\frac{L-x}{x} < \frac{1}{4} \Rightarrow 4L - 4x < x \Rightarrow 5x > 4L \Rightarrow x > \frac{4L}{5}$$

Valid Region

$$(0, \frac{L}{5}) \cup (\frac{4L}{5}, L)$$

Total length:

$$\frac{L}{5} + \frac{L}{5} = \frac{2L}{5}$$

Final Probability

$$P = \frac{2L/5}{L} = \frac{2}{5}$$

Q2. Bayesian Classification with Normal Distributions

Given

- $X|W \sim N(4, 4) \Rightarrow \sigma_1 = 2$

- $X|B \sim N(6,9) \Rightarrow \sigma_2 = 3$
- $P(B) = \alpha, \quad P(W) = 1 - \alpha$
- Observation: $X = 5$

Condition for Equal Error

$$P(B|X=5) = P(W|X=5) = \frac{1}{2}$$

Using Bayes' Theorem

$$\frac{\alpha f(5|B)}{\alpha f(5|B) + (1-\alpha) f(5|W)} = \frac{1}{2}$$

Normal Densities

$$f(5|B) = \frac{1}{3\sqrt{2\pi}} e^{-1/18}, \quad f(5|W) = \frac{1}{2\sqrt{2\pi}} e^{-1/8}$$

Substitution

$$\frac{3\alpha}{3\alpha + 2(1-\alpha)e^{-5/72}} = \frac{1}{2}$$

Solving

$$6\alpha = 3\alpha + 2(1-\alpha)e^{-5/72}$$

$$3\alpha = 2(1-\alpha)(0.93)$$

$$3\alpha = 1.86 - 1.86\alpha$$

$$4.86\alpha = 1.86 \Rightarrow \alpha \approx 0.3827$$

Result

$$\alpha \approx 0.38$$

Q3. Optimal Location of Fire Station

(a) Uniform Distribution

Let:

$$X \sim \text{Uniform}(0, A), \quad f_X(x) = \frac{1}{A}$$

Minimize:

$$E[|X - a|]$$

Expectation

$$E[|X - a|] = \int_0^a (a - x)\frac{1}{A}dx + \int_a^A (x - a)\frac{1}{A}dx$$

Derivative

$$\frac{d}{da}\left(\frac{2a^2 - 2aA + A^2}{2A}\right) = \frac{1}{2A}(4a - 2A)$$

Set to zero:

$$4a = 2A \Rightarrow a = \frac{A}{2}$$

Result

Station at midpoint.

(b) Exponential Distribution

$$X \sim \text{Exponential}(\lambda), \quad f_X(x) = \lambda e^{-\lambda x}$$

Derivative of Expectation

$$\frac{d}{da}E[|X-a|] = 1 - 2e^{-\lambda a} = 0$$

Solve

$$2e^{-\lambda a} = 1 \Rightarrow e^{-\lambda a} = \frac{1}{2}$$

$$-\lambda a = \ln\left(\frac{1}{2}\right) = -\ln 2 \Rightarrow a = \frac{\ln 2}{\lambda}$$

Q4. Conditional Survival of Mixed Exponential

Given

Two battery types:

- Type 1: λ_1, probability p_1
- Type 2: λ_2, probability p_2

Total Probability

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

$$P(X > t+s) = p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}$$

Conditional Probability

$$P(X > t+s | X > t) = \frac{p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}$$

Q5. Conditional Distribution of Poisson Variables

Given

$$X \sim \text{Poisson}(\lambda_1), \quad Y \sim \text{Poisson}(\lambda_2)$$

Required

$$P(X = k | X + Y = n)$$

Using Independence

$$= \frac{P(X = k)P(Y = n - k)}{P(X + Y = n)}$$

Substitute PMFs

$$= \frac{e^{-\lambda_1}\lambda_1^k/k! \cdot e^{-\lambda_2}\lambda_2^{n-k}/(n-k)!}{e^{-(\lambda_1+\lambda_2)}(\lambda_1 + \lambda_2)^n/n!}$$

Final Form

$$= \binom{n}{k}\left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n-k}$$

Q6. Transformation of Random Variable

Given

$$X \sim \text{Uniform}[-2, 2], \quad f_X(x) = \frac{1}{4}$$

Piecewise:

$$g(x) = \begin{cases} x, & [-2, -1] \\ 0, & (-1, 1) \\ x, & [1, 2] \end{cases}$$

CDF $F_Y(y)$

Piecewise evaluation:

- $y < -2$: 0
- $-2 \leq y < -1$: $\frac{y+2}{4}$

- $-1 \leq y < 0$: $\frac{1}{4}$
- At $y = 0$: jump of $\frac{1}{2} \Rightarrow F_Y(0) = \frac{3}{4}$
- $0 < y < 1$: $\frac{3}{4}$
- $1 \leq y \leq 2$: $\frac{y+2}{4}$
- $y > 2$: 1

PDF

$$f_Y(y) = \begin{cases} \frac{1}{4}, & -2 < y < -1 \\ \frac{1}{4}, & 1 < y < 2 \\ 0, & \text{otherwise} \end{cases}$$

With point mass:

$$P(Y = 0) = \frac{1}{2}$$

Q7. Stock Model (Binomial Approximation)

Final Price

$$S_{1000} = S_0 u^k d^{1000-k}$$

Condition:

$$u^k d^{1000-k} \geq 1.30$$

Taking Log

$$k \ln u + (1000 - k) \ln d \geq \ln(1.30)$$

Solve for k

$$k \geq \frac{\ln(1.30) - 1000 \ln d}{\ln u - \ln d} \approx 469.4 \Rightarrow k \geq 470$$

Distribution

$$k \sim \text{Binomial}(1000, 0.52)$$

Final Probability

$$P(k \geq 470) = \sum_{k=470}^{1000} \binom{1000}{k} (0.52)^k (0.48)^{1000-k}$$

Q8. Exponential Distribution

PDF

$$f(x) = \frac{1}{2} e^{-x/2}, \quad x \geq 0$$

(a)

$$P(X > 2) = e^{-1}$$

(b)

$$P(X \geq 10 | X > 9) = \frac{e^{-5}}{e^{-9/2}} = e^{-1/2}$$

Q9. Transformation $Z = \sqrt{X^2 + Y^2}$

CDF

$$F_Z(z) = P(X^2 + Y^2 \leq z^2)$$

Joint PDF

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

Region

$$-z \le x \le z, \quad -\sqrt{z^2 - x^2} \le y \le \sqrt{z^2 - x^2}$$

CDF Integral

$$F_Z(z) = \int_{-z}^{z} \int_{-\sqrt{z^2-x^2}}^{\sqrt{z^2-x^2}} f_X(x) f_Y(y)\, dy\, dx$$

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$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Using Leibniz rule:

$$f_Z(z) = z \int_{-z}^{z} \frac{f_X(x) \left[f_Y(\sqrt{z^2 - x^2}) + f_Y(-\sqrt{z^2 - x^2}) \right]}{\sqrt{z^2 - x^2}} dx$$

Q10. Joint Distribution of M_n and M_{n+1}

Definitions

$$M_n = \max(X_1, \ldots, X_n), \quad M_{n+1} = \max(M_n, X_{n+1})$$

CDF of M_n

$$P(M_n \le x) = F(x)^n$$

Joint CDF

Case 1: $x \le y$

$$P(M_n \le x, M_{n+1} \le y) = F(x)^n F(y)$$

Case 2: $x > y$

$$P(M_n \le x, M_{n+1} \le y) = F(y)^{n+1}$$

Final Result

$$P(M_n \le x, M_{n+1} \le y) = \begin{cases} F(x)^n F(y), & x \le y \\ F(y)^{n+1}, & x > y \end{cases}$$

If you want, I can convert this into **clean handwritten-style notes**, **LaTeX PDF**, or **short revision cheat sheet** for exams.